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Quasi-norm error bounds for the finite element approximation of a non-Newtonian flow

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Summary. We consider the finite element approximation of a non-Newtonian flow, where the viscosity obeys a general law including the Carreau or power law. For sufficiently regular solutions we prove energy type error bounds for the velocity and pressure. These bounds improve on existing results in the literature. A key step in the analysis is to prove abstract error bounds initially in a quasi-norm, which naturally arises in degenerate problems of this type.

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1. Introduction

Let Ω be a bounded open set in $\mathbb{R}^2$ with a Lipschitz boundary Γ. Given $\mathbf{f} \in [L^2(\Omega)]^2$ and a positive function $k \in C(0,\infty)$, we consider the following boundary value problem:

($\mathscr{P}$) Find $(\mathbf{u}, p)$ such that

$$-\sum_{j=1}^{2} \partial(k(|D(\mathbf{u})|)D_{ij}(\mathbf{u}))/\partial x_j + \partial p/\partial x_i = f_i \quad \text{in} \quad \Omega \quad 1 \le i \le 2\ , \tag{1.1a}$$

$$\operatorname{div} \mathbf{u} = 0 \quad \text{in } \Omega\ , \tag{1.1b}$$

$$\mathbf{u} = \mathbf{0} \text{ on } \Gamma \quad \text{and} \quad \int_\Omega p = 0\ ; \tag{1.1c}$$

where $D(\mathbf{u})$ is the rate of deformation tensor with entries

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(1.2a) $$D_{ij}(\mathbf{u}) \equiv \tfrac{1}{2}(\partial u_i/\partial x_j + \partial u_j/\partial x_i)$$

and

(1.2b) $$|D(\mathbf{u})|^2 \equiv \sum_{i,j=1}^{2} [D_{ij}(\mathbf{u})]^2 \ .$$

Throughout $|\cdot|$ denotes the Euclidean (Frobenius) matrix norm; that is for $X \in \mathbb{M}_n$, $n \times n$ real matrices, $|X| \equiv \left[\sum_{i,j=1}^{n}(X_{ij}^2)\right]^{1/2}$. The above models the steady isothermal flow of an incompressible fluid, where the nonlinear inertial terms have been neglected. **f** is the applied body forces, **u** the velocity, p the pressure and k the viscosity of the fluid. For a Newtonian fluid the viscosity k is assumed to be constant, $k \equiv k_0$. I this case (1.1) is the much studied linear Stokes problem and ($\mathscr{P}$) can be written more conveniently as

($\mathscr{P}_S$) Find $(\mathbf{u}, p)$ such that

(1.3a) $$-\tfrac{1}{2}k_0 \Delta \mathbf{u} + \text{ grad } p = \mathbf{f} \quad \text{in} \quad \Omega \ ,$$

(1.3b) $$\text{div } \mathbf{u} = 0 \quad \text{in} \quad \Omega \ ,$$

(1.3c) $$\mathbf{u} = \mathbf{0} \text{ on } \Gamma \quad \text{and} \quad \int_\Omega p = 0 \ .$$

The well-posedness of ($\mathscr{P}_S$) and its finite element approximation are well-established, see Girault and Raviart (1986) (abbreviated to G&R throughout this paper).

Baranger and Najib (1990) (abbreviated to B&N throughout this paper) have extended these well established finite element approximation of ($\mathscr{P}_S$) to the non-linear problem ($\mathscr{P}$), when the viscosity k obeys either

(a) the Carreau law

(1.4a) $$k(t) \equiv k_\infty + (k_0 - k_\infty)(1 + \lambda t^2)^{r-2/2}$$

or (b) the power law

(1.4b) $$k(t) \equiv k_0 t^{r-2} \ ;$$

where $k_0 > k_\infty \geq 0$, $\lambda > 0$ and $r \in (1, \infty)$. We note that the choice $r = 2$ in either (1.4a) or (1.4b) leads to $k \equiv k_0$, Newtonian flow, and hence (1.3). (1.1) with either (1.4a) or (1.4b) models a pseudo-plastic fluid if $r \in (1, 2)$ and a dilatant fluid if $r > 2$. Extensions and improvements on the error bounds of B&N have recently appeared in Sandri (1993) and in Barrett and Liu (1993a) (abbreviated to B&L throughout this paper). It is the purpose of this paper to further extend and improve on these bounds; and also to simplify the analysis. Below we review the present state of the error analysis. We assume that the assumptions (H1) and (H2) hold, see Sect. 4, and the appropriate Babuska/Brezzi inf-sup conditions hold on the approximation spaces for the velocity and pressure, see Sects. 2 and 4.

Firstly we consider the Carreau law. If $r \in (1, 2]$ and $k_\infty > 0$ optimal error bounds for the velocity in $[W^{1,2}(\Omega)]^2$ and pressure in $L^2(\Omega)$ are easily established, see B&N and also B&L. If $r \in (1, 2]$ and $k_\infty = 0$ an optimal error bound, h^j, for the

velocity in $[W^{1,r}(\Omega)]^2$ and an $h^{j(r-1)}$ error bound for the pressure in $L^{r'}(\Omega)$, where $r' = r/(r-1)$, are established in B&L. These are superior to the bounds in B&N and Sandri (1993). The analysis in this paper will yield an optimal error bound for the pressure in $L^2(\Omega)$. If $r \in [2, \infty)$ optimal error bounds for the velocity in $[W^{1,2}(\Omega)]^2$ and pressure in $L^{r'}(\Omega)$ are established in B&L. The bound on the pressure required a quasi-uniform mesh in order for an inverse assumption to hold. This restriction is removed by the analysis in this paper. Finally we note that a similar law, the non-degenerate "power law" $k(t) = k_\infty + (k_0 - k_\infty)t^{r-2}$ with $k_0 > k_\infty > 0$, is analysed in Du and Gunzburger (1990) for $r \in [2, \infty)$.

We now consider the power law. If $r \in (1, 2]$ an $h^{jr/2}$ error bound for the velocity in $[W^{1,r}(\Omega)]^2$ and an $h^{jr(r-1)/2}$ error bound for the pressure in $L^{r'}(\Omega)$ are established in B&N, see also B&L. The above bound on the pressure is improved to $h^{j(r-1)}$ in Sandri (1993). For the P_1 flux – P_0 element of Fortin (1981), which satisfies assumptions (H1) and (H2) with $j = 1$, B&L proved an optimal error bound, Ch, for the velocity in $[W^{1,r}(\Omega)]^2$ and an h^{r-1} error bound for the pressure in $L^{r'}(\Omega)$ under stronger regularity requirements on the solution $\mathbf{u}$ than for the above results with $j = 1$. This bound on the pressure is improved to $h^{2(r-1)/r}$ in this paper. If $r \in [2, \infty)$ then $h^{j/(r-1)}$ error bounds for the velocity in $[W^{1,r}(\Omega)]^2$ and pressure in $L^{r'}(\Omega)$ are established in B&L. The bound on the pressure is improved to $h^{jr/[2(r-1)]}$ in Sandri (1993).

In this paper we will analyse a more general viscosity law, showing how the standard error bounds deteriorate as the law becomes more degenerate. The approach adopted in this paper is substantially different with that used in B&N and B&L, where the associated minimisation problems are exploited. Like Du and Gunzburger (1990) and Sandri (1993) the present analysis can be applied when the nonlinear inertial terms are included, and even to the time dependent case, by adapting the techniques used in Barrett and Liu (1993b) and Liu and Barrett (1993) for degenerate quasilinear parabolic problems. A key step in the analysis is to prove abstract error bounds initially in a quasi-norm, which naturally arises in degenerate problems of this type. All the above results for the Carreau and power law, including the non-degenerate version, with the stated improvements can be obtained from these abstract bounds. Moreover, in many cases for a quasi-uniform mesh these error bounds are shown to hold in stronger norms than stated above. We believe that the introduction of this quasi-norm not only leads to a very clean and simple abstract theory, but is also an essential ingredient in proving sharp error bounds.

The layout of this paper is as follows. In the next section we introduce our assumptions on the viscosity law $k(\cdot)$ and establish some important inequalities. In addition we introduce an abstract finite element approximation of (1.1) and establish well-posedness. In Sect. 3 we prove some abstract error bounds, initially in a quasi-norm and then in standard Sobolev spaces. Finally in Sect. 4 we firstly establish error bounds under some general hypotheses and then study in more detail the P_1 flux – P_0 scheme of Fortin (1981). Throughout this paper we adopt the standard notation $W^{\nu,s}(\Omega)$ for Sobolev spaces with norm $\|\cdot\|_{W^{\nu,s}(\Omega)}$ and semi-norm $|\cdot|_{W^{\nu,s}(\Omega)}$. We set

$$W_0^{1,\nu}(\Omega) \equiv \{w \in W^{1,\nu}(\Omega) : w = 0 \text{ on } \partial\Omega\}$$

and

$$L_0^{\nu}(\Omega) \equiv \left\{ w \in L^{\nu}(\Omega) : \int_\Omega w = 0 \right\} .$$

Similarly, we adopt the standard notation $C^{m,\alpha}(\overline{\Omega})$ for Hölder spaces with norm $\|\cdot\|_{C^{m,\alpha}(\overline{\Omega})}$. We shall extend these norms in the natural way to vector functions, which are written in bold-face characters. Throughout C and C_i denote generic constants independent of h. In addition $C(t_1,\ldots,t_m)$ denotes a constant depending on the non-negative parameters t_i, $i = 1 \to m$, such that $C(t_1,\ldots,t_m) \leq C$ if $t_i \leq C$ $i = 1 \to m$.

2. Preliminaries

Before considering the weak formulation of ($\mathscr{P}$) we first state some very important inequalities which have been recently used to obtain error bounds for the finite element approximation of degenerate problems, see Liu and Barrett (1992, 1993).

Assumptions (A): We assume that $k \in C(\overline{\Omega} \times (0,\infty))$ and that there exist constants $r \in (1,\infty)$, $\alpha \in [0,1]$ and $\varepsilon, C, M > 0$ such that for all $\mathbf{x} \in \overline{\Omega}$

(A1) $$k(\mathbf{x},t) \leq C[t^\alpha(1+t)^{1-\alpha}]^{r-2} \text{ for all } t \geq 0$$

and

$$|k(\mathbf{x},t)t - k(\mathbf{x},s)s| \leq C|t-s|[(t+s)^\alpha(1+t+s)^{1-\alpha}]^{r-2}$$
$$\text{for all } s,t > 0 \text{ satisfying } |s/t - 1| \leq \varepsilon$$

(A2) $$k(\mathbf{x},t)t - k(\mathbf{x},s)s \geq M(t-s)[(t+s)^\alpha(1+t+s)^{1-\alpha}]^{r-2} \text{ for all } t \geq s \geq 0\ .$$

Throughout this paper, for notational simplicity, we will denote $k(\cdot,t)$ by $k(t)$.

Remark 2.1. Many viscosity laws met in practical problems satisfy the Assumptions (A). For example: the Carreau law, (1.4a),

$$k(t) \equiv k_\infty + (k_0 - k_\infty)(1+\lambda t^2)^{(\theta-2)/2} \text{ with } k_0 > k_\infty \geq 0\ , \quad \lambda > 0 \text{ and } \theta \in (1,\infty)\ ,$$
$$\Rightarrow \alpha = 0 \text{ with } r = \theta \text{ if } k_\infty = 0 \text{ and } r = 2 \text{ if } \theta \in (1,2] \text{ and } k_\infty > 0\ ;$$

the generalised power law, choosing $\mu = 1$ yields the power law (1.4b),

$$k(t) \equiv k_0[t^\mu(1+t)^{1-\mu}]^{\theta-2} \text{ with } k_0 > 0\ , \mu \in [0,1] \text{ and } \theta \in (1,\infty)$$
$$\Rightarrow \alpha = \mu \text{ and } r = \theta\ ;$$

and the non-degenerate "power law"

$$k(t) \equiv k_\infty + (k_0 - k_\infty)t^{\theta-2} \text{ with } k_0 > k_\infty > 0 \text{ and } \theta \in [2,\infty)\ ,$$
$$\Rightarrow \alpha = 0 \text{ and } r = \theta\ .$$

The parameter α in (A) measures the degree of degeneracy in $k(\cdot)$ for a given r, in that the closer α is to one the more degenerate $k(\cdot)$ is. This separation of $(\cdot)$ into its degenerate and non-degenerate parts plays a crucial role in the inequalities that follow and the subsequent error analysis.

Throughout this section we adopt the following notation: $\mathbb{M}_n$ is the set of $n \times n$ real matrices, $(\cdot,\cdot)$ denotes the inner product on $\mathbb{M}_n$; that is, for $X, Y \in \mathbb{M}_n$ $(X,Y) \equiv \sum_{i,j=1}^n X_{ij}Y_{ij}$ and $|X| \equiv [(X,X)]^{1/2}$.

Lemma 2.1. *Let k satisfy the Assumptions* (A1) *for $r \in (1,\infty)$ and $\alpha \in [0,1]$. Then for all $X, Y \in \mathbb{M}_n$ ($n \geq 1$) and $\delta \geq 0$ we have that*

(2.1a) $|k(|X|)X - k(|Y|)Y| \le C|X-Y|^{1-\delta}[(|X|+|Y|)^{\alpha}(1+|X|+|Y|)^{1-\alpha}]^{r-2+\delta}$.

Let k satisfy the Assumptions (A2) *for $r \in (1,\infty)$ and $\alpha \in [0,1]$. Then for all $X, Y \in \mathbb{M}_n$ ($n \ge 1$) and $\delta \ge 0$ we have that*

$$\begin{aligned}&(k(|X|)X - k(|Y|)Y, X-Y)\\ &\quad \ge M|X-Y|^{2+\delta}[(|X|+|Y|)^{\alpha}(1+|X|+|Y|)^{1-\alpha}]^{r-2-\delta}\ .\end{aligned} \tag{2.1b}$$

Proof. We extend the proof given in Liu & Barrett (1992); where $(\cdot,\cdot)$ is the inner product on $\mathbb{R}^n$, X, $Y \in \mathbb{R}^n$ and techniques similar to those in Chow (1989) are applied. We first prove (2.1a) for $\delta = 0$. For any $(X,Y) \in \mathbb{M}_n \times \mathbb{M}_n$ let

$$F(X,Y) \equiv |k(|X|)X - k(|Y|)Y|/\{|X-Y|[(|X|+|Y|)^{\alpha}(1+|X|+|Y|)^{1-\alpha}]^{r-2}\}\ .$$

We wish to prove that F is bounded. For any $\varepsilon > 0$ F is continuous on $D_\varepsilon \equiv \{(X,Y) \in \mathbb{M}_n \times \mathbb{M}_n : |X-Y| \ge \varepsilon \text{ and } |X|+|Y| \le 1/\varepsilon\}$. Furthermore it follows for all $(X,Y) \in \mathbb{M}_n \times \mathbb{M}_n$ that

$$F(X,Y) = F(Y,X) \quad \text{and} \quad F(0,Y) \le C\ . \tag{2.2}$$

Therefore without loss of generality we can suppose that $X, Y \ne 0$. Let $b(X) \equiv [|X|^{\alpha}(1+|X|)^{1-\alpha}]^{r-2}$. Then F can be rewritten as

$$F(X,Y) \equiv \frac{|k(|X|)X/b(X)|X| - [k(|Y|)Y/b(Y)|Y|][b(Y)|Y|/b(X)|X|]|}{|X/|X| - Y/|X||[(1+|Y|/|X|)^{\alpha}(1+|Y|/(1+|X|))^{1-\alpha}]^{r-2}}\ .$$

We can further assume from (2.2) that $|Y|/|X| \le 1$. It follows that $F(X,Y)$ will be bounded on $\mathbb{M}_n \times \mathbb{M}_n$ if $Y/|X|$ does not tend to $E \equiv X/|X|$ as the function $X \to k(|X|)/b(X)$ is bounded above. We now show that $\overline{\lim}_{|Y|/|X|\to 1} F(X,Y) \le C$.

If $1-\varepsilon \le |Y|/|X| \le 1$ for some $\varepsilon \in (0,1)$ then there exists a constant C such that $|X||Y| \le (|X|+|Y|)^2 \le C|X||Y|$. Then it follows, since $|X||Y| - (X,Y) \le |X-Y|^2$, that

$$\begin{aligned}|k(|X|)X - k(|Y|)Y|^2 &= (k(|X|)|X| - k(|Y|)|Y|)^2\\ &\quad + 2k(|X|)k(|Y|)(|X||Y| - (X,Y))\\ &\le C(|X|-|Y|)^2[(|X|+|Y|)^{\alpha}(1+|X|+|Y|))^{1-\alpha}]^{2(r-2)}\\ &\quad + C|X-Y|^2\{(|X||Y|)^{\alpha}[(1+|X|)(1+|Y|)]^{1-\alpha}\}^{r-2}\\ &\le C|X-Y|^2[(|X|+|Y|)^{\alpha}(1+|X|+|Y|)^{1-\alpha}]^{2(r-2)}\ .\end{aligned}$$

Consequently one has that (2.1a) holds with $\delta = 0$. On the other hand for all $(X,Y) \in \mathbb{M}_n \times \mathbb{M}_n$ and $\delta \ge 0$ we have that

$$\begin{aligned}&|X-Y|[(|X|+|Y|)^{\alpha}(1+|X|+|Y|)^{1-\alpha}]^{r-2}\\ &\le |X-Y|^{1-\delta}[(|X|+|Y|)^{\alpha}(1+|X|+|Y|)^{1-\alpha}]^{r-2+\delta}\ ;\end{aligned}$$

that is, (2.1a) holds for any $\delta \ge 0$.

Similarly (2.1b) for the case $\delta = 0$ holds since we have for $M_i > 0$ that

$$
\begin{aligned}
&\mathbf{(}k(|X|)X - k(|Y|)Y, X - Y\mathbf{)}\\
&= (k(|X|)|X| - k(|Y|)|Y|)(|X| - |Y|) + (k(|X|) + k(|Y|))(|X||Y| - \mathbf{(}X, Y\mathbf{)})\\
&\geq M(|X| - |Y|)^2[(|X| + |Y|)^\alpha(1 + |X| + |Y|)^{1-\alpha}]^{r-2}\\
&\quad + M\{[|X|^\alpha(1 + |X|)^{1-\alpha}]^{r-2} + [|Y|^\alpha(1 + |Y|)^{1-\alpha}]^{r-2}\}(|X||Y| - \mathbf{(}X, Y\mathbf{)})\\
&\geq M_1|X - Y|^2[(|X| + |Y|)^\alpha(1 + |X| + |Y|)^{1-\alpha}]^{r-2} ,
\end{aligned}
$$

where we have noted for all $s, t \geq 0$ that

$$
[s^\alpha(1 + s)^{1-\alpha}]^{r-2} + [t^\alpha(1 + t)^{1-\alpha}]^{r-2} \geq M_2[(s + t)^\alpha(1 + s + t)^{1-\alpha}]^{r-2} . \tag{2.3}
$$

Clearly (2.3) holds for $r \in (1, 2]$ with $M_2 = 1$. For $r \in (2, \infty)$, (2.3) follows from noting that there exist constants M_3 and $M_4(r)$ such that

$$
(s + t)^\alpha(1 + s + t)^{1-\alpha} \leq M_3[s^\alpha(1 + s)^{1-\alpha} + t^\alpha(1 + t)^{1-\alpha}]
$$

and

$$
(s + t)^{r-2} \leq M_4[s^{r-2} + t^{r-2}] .
$$

In addition for all $(X, Y) \in \mathbb{M}_n \times \mathbb{M}_n$ and $\delta \geq 0$ we have that

$$
\begin{aligned}
&|X - Y|^2[(|X| + |Y|)^\alpha(1 + |X| + |Y|)^{1-\alpha}]^{r-2}\\
&\geq |X - Y|^{2+\delta}[(|X| + |Y|)^\alpha(1 + |X| + |Y|)^{1-\alpha}]^{r-2-\delta}
\end{aligned}
$$

and hence (2.1b) holds for all $\delta \geq 0$. □

Remark 2.2. Under similar assumptions Chow (1989) has proved the following versions of (2.1a&b) for $X, Y \in \mathbb{R}^n$, where $\mathbf{(}\cdot, \cdot\mathbf{)}$ is the inner product on $\mathbb{R}^n$: (2.1a) with $\delta = 2 - r$ for $r \in (1, 2]$ and $\alpha = 1$; (2.1a) with $\delta = 0$ and $\alpha = 0$ for $r \in [2, \infty)$; (2.1b) with $\delta = 0$ and $\alpha = 0$ for $r \in (1, 2]$; (2.1b) with $\delta = r - 2$ for $r \in [2, \infty)$ and $\alpha = 1$. With these one can establish some error bounds for the finite element approximation of ($\mathscr{P}$) as in B&N. Sandri (1993) has improved on the results of Chow (1989) by taking into account the degenerate ($\alpha = 1$), non-degenerate ($\alpha = 0$) nature of k. However, the above generalizations are a key ingredient to proving superior error bounds.

Another key ingredient is the following result, which is a generalisation of Lemma 2.3 in Barrett & Liu (1993b). This lemma allows us to avoid the use of the associated minimisation problems, as exploited in B&N and B&L. Therefore the present analysis can be applied when the nonlinear inertial terms are included, and even to the unsteady version.

Lemma 2.2. *For all $r \in (1, \infty)$, $\alpha \in [0, 1]$, a, σ_1, $\sigma_2 \geq 0$ and for all $\varepsilon \in (0, 1]$*

$$
\begin{aligned}
&[(a + \sigma_1)^\alpha(1 + a + \sigma_1)^{1-\alpha}]^{r-2}\sigma_1\sigma_2 \leq \varepsilon[(a + \sigma_1)^\alpha(1 + a + \sigma_1)^{1-\alpha}]^{r-2}\sigma_1^2\\
&\quad + \varepsilon^{-\gamma}[(a + \sigma_2)^\alpha(1 + a + \sigma_2)^{1-\alpha}]^{r-2}\sigma_2^2 ,
\end{aligned} \tag{2.4}
$$

where $\gamma = \max\{1, r - 1\}$.

Proof. Clearly (2.4) holds if $\sigma_1\sigma_2 = 0$, so we can assume that $\sigma_1, \sigma_2 > 0$. It follows for $0 < \sigma_1 \leq \sigma_2$ that

$$
[(a + \sigma_1)^\alpha(1 + a + \sigma_1)^{1-\alpha}]^{r-2}\sigma_1\sigma_2 \leq [(a + \sigma_2)^\alpha(1 + a + \sigma_2)^{1-\alpha}]^{r-2}\sigma_2^2 \tag{2.5}
$$

since $[(a+\sigma)^{\alpha}(1+a+\sigma)^{1-\alpha}]^{r-2}\sigma$ is increasing for $\sigma \geq 0$. Assume now that $0 < \sigma_2 \leq \sigma_1$.

If $r \leq 2$ then

$$\varepsilon(\sigma_1/\sigma_2) + \varepsilon^{-1}(\sigma_2/\sigma_1)\{[(a+\sigma_2)/(a+\sigma_1)]^{\alpha}[(1+a+\sigma_2)/(1+a+\sigma_1)]^{1-\alpha}\}^{r-2}$$
$$\geq \varepsilon(\sigma_1/\sigma_2) + \varepsilon^{-1}(\sigma_2/\sigma_1) \geq 2 \,. \tag{2.6}$$

If $r \geq 2$ then

$$\varepsilon(\sigma_1/\sigma_2) + \varepsilon^{-\gamma}(\sigma_2/\sigma_1)\{[(a+\sigma_2)/(a+\sigma_1)]^{\alpha}[(1+a+\sigma_2)/(1+a+\sigma_1)]^{1-\alpha}\}^{r-2}$$
$$\geq \varepsilon(\sigma_1/\sigma_2) + (\varepsilon^{-1}\sigma_2/\sigma_1)^{r-1} = \xi + \xi^{1-r} \geq 1 \,, \tag{2.7}$$

where $\xi \equiv \varepsilon(\sigma_1/\sigma_2)$. The inequality (2.4) then follows from (2.5–2.7). □

Below we introduce the direct and mixed weak formulations of ($\mathscr{P}$) and their corresponding discrete approximations. We will assume that k satisfies Assumptions (A). Let

$$J(\mathbf{v}) \equiv \int_{\Omega} \left[\int_0^{|D(\mathbf{v})|} k(t)t \; dt \right] - (\mathbf{f}, \mathbf{v}) \,, \tag{2.8a}$$

where

$$(\mathbf{f}, \mathbf{v}) \equiv \sum_{j=1}^{2} (f_j, v_j) \quad \text{and} \quad (f_j, v_j) \equiv \int_{\Omega} f_j v_j \,. \tag{2.8b}$$

We have from Assumptions (A) that $J(\cdot)$ is well defined on $[W^{1,r}(\Omega)]^2$. In addition we note the Poincaré and generalised Korn inequality, see Mosolov and Mjasnikov (1971), for $\nu \in (1,\infty)$ and for all $\mathbf{w} \in [W_0^{1,\nu}(\Omega)]^2$

$$|\mathbf{w}|_{D^{\nu}(\Omega)} \equiv \left\{ \int_{\Omega} |D(\mathbf{w})|^{\nu} \right\}^{1/\nu} \geq C_1 |\mathbf{w}|_{W^{1,\nu}(\Omega)} \geq C_2 \|\mathbf{w}\|_{W^{1,\nu}(\Omega)} \,. \tag{2.9}$$

We set $\mathbf{X} \equiv [W_0^{1,r}(\Omega)]^2$ and $\|\cdot\|_{\mathbf{X}} \equiv |\cdot|_{W^{1,r}(\Omega)}$. Let $\langle \cdot, \cdot \rangle$ denote the duality pairing between $\mathbf{X}^*$ and $\mathbf{X}$. It is easy to check that $J(\cdot)$ is Gateaux differentiable on $\mathbf{X}$ with

$$\langle J'(\mathbf{w}_1), \mathbf{w}_2 \rangle \equiv \langle A\mathbf{w}_1, \mathbf{w}_2 \rangle - (\mathbf{f}, \mathbf{w}_2) \quad \forall \mathbf{w}_1, \mathbf{w}_2 \in \mathbf{X} \,; \tag{2.10a}$$

where $A : \mathbf{X} \to \mathbf{X}^*$ is such that

$$\langle A\mathbf{w}_1, \mathbf{w}_2 \rangle \equiv \int_{\Omega} \mathbf{(}k(|D(\mathbf{w}_1)|)D(\mathbf{w}_1), D(\mathbf{w}_2)\mathbf{)} \,. \tag{2.10b}$$

In addition $J'(\cdot)$ is strictly monotone, and hence $J(\cdot)$ is strictly convex on $\mathbf{X}$; see Lemma 2.1 above. Furthermore $J(\cdot)$ is coercive on $\mathbf{X}$. Setting

$$\mathbf{V} \equiv \{\mathbf{v} \in \mathbf{X} : \; \text{div } \mathbf{v} = 0 \text{ in } \Omega\} \,, \tag{2.11}$$

it follows that there exists a unique solution to the minimization problem:

($\mathscr{O}$) Find $\mathbf{u} \in \mathbf{V}$ such that

$$J(\mathbf{u}) \leq J(\mathbf{v}) \quad \forall \mathbf{v} \in \mathbf{V} \,. \tag{2.12}$$

Furthermore ($\mathscr{Q}$) is equivalent to ($\mathscr{R}$), its Euler equation, where

($\mathscr{R}$) Find $\mathbf{u} \in \mathbf{V}$ such that

$$\langle A\mathbf{u}, \mathbf{v}\rangle = (\mathbf{f}, \mathbf{v}) \quad \forall \mathbf{v} \in \mathbf{V} \ . \tag{2.13}$$

One can view ($\mathscr{R}$) as the direct weak formulation of ($\mathscr{P}$). From (2.13) with $\mathbf{v} \equiv \mathbf{u}$ it follows from (2.10b) and (2.1b) that

$$\|\mathbf{u}\|_{D^r(\Omega)} \leq C(\|\mathbf{f}\|_{L^2(\Omega)}) \ . \tag{2.14}$$

We now consider the mixed weak formulation of ($\mathscr{P}$). We set

$$M \equiv L_0^{r'}(\Omega) \quad \text{and} \quad \|\cdot\|_M \equiv \|\cdot\|_{L^{r'}(\Omega)} \ , \tag{2.15}$$

with $r' = r/(r-1) \Rightarrow (1/r) + (1/r') = 1$. Then the mixed weak formulation of ($\mathscr{P}$) is

($\mathscr{S}$) Find $(\mathbf{u}, p) \in \mathbf{X} \times M$ such that

$$\langle A\mathbf{u}, \mathbf{w}\rangle - (p, \ \text{div}\ \mathbf{w}) = (\mathbf{f}, \mathbf{w}) \quad \forall \mathbf{w} \in \mathbf{X} \tag{2.16a}$$

$$(\text{div}\ \mathbf{u}, q) = 0 \quad \forall q \in M \ . \tag{2.16b}$$

It follows from taking $\mathbf{w} \in \mathbf{V} \subset \mathbf{X}$ in (2.16a) that $\mathbf{u}$, the solution of ($\mathscr{Q}$) $\equiv$ ($\mathscr{R}$) is the unique $\mathbf{u}$ that solves ($\mathscr{S}$). For the well-posedness of p in ($\mathscr{S}$) we require the Babuska/Brezzi inf-sup condition,

$$\inf_{q \in M} \sup_{\mathbf{w} \in \mathbf{X}} (q, \ \text{div}\ \mathbf{w})/[\|q\|_M \ \|\mathbf{w}\|_{\mathbf{X}}] \geq \beta > 0 \ , \tag{2.17a}$$

to hold. Amrouche and Girault (1990), see Remark 3.2 and Corollary 3.1, prove the following result: For any $\nu \in (1, \infty)$ there exists a positive constant $\beta(\nu)$ such that

$$\inf_{q \in L_0^{\nu'}(\Omega)} \sup_{\mathbf{w} \in [W_0^{1,\nu}(\Omega)]^2} (q, \ \text{div}\ \mathbf{w})/[\|q\|_{L^{\nu'}(\Omega)} \|\mathbf{w}\|_{W^{1,\nu}(\Omega)}] \geq \beta(\nu) > 0 \ , \tag{2.17b}$$

where $\nu' = \nu/(\nu - 1)$. Hence we have that (2.17a) holds, which in turn implies the existence of a unique solution $(\mathbf{u}, p)$ to ($\mathscr{S}$).

Let $\mathbf{X}_h$ and M_h be two finite dimensional spaces such that

$$\mathbf{X}_h \subset \mathbf{X} \cap [W^{1,\infty}(\Omega)]^2 \quad \text{and} \quad M_h \subset M \cap L^\infty(\Omega) \ . \tag{2.18}$$

The inclusion of $\mathbf{X}_h$ in $[W^{1,\infty}(\Omega)]^2$ and M_h in $L^\infty(\Omega)$ is exploited in some arguments later, and as these hold for all practical choices of $\mathbf{X}_h$ and M_h, we state it as an assumption from the outset. We then set

$$\mathbf{V}_h \equiv \{\mathbf{v}_h \in \mathbf{X}_h : (\text{div}\ \mathbf{v}_h, q_h) = 0 \ \forall q_h \in M_h\} \ . \tag{2.19}$$

The corresponding approximations of problems ($\mathscr{Q}$), ($\mathscr{R}$) and ($\mathscr{S}$) are then:

($\mathscr{Q}_h$) Find $\mathbf{u}_h \in \mathbf{V}_h$ such that

$$J(\mathbf{u}_h) \leq J(\mathbf{v}_h) \quad \forall \mathbf{v}_h \in \mathbf{V}_h \ . \tag{2.20}$$

($\mathscr{R}_h$) Find $\mathbf{u}_h \in \mathbf{V}_h$ such that

(2.21) $$\langle A\mathbf{u}_h, \mathbf{v}_h\rangle = (\mathbf{f}, \mathbf{v}_h) \quad \forall \mathbf{v}_h \in \mathbf{V}_h \ .$$

($\mathscr{S}_h$) Find $(\mathbf{u}_h, p_h) \in \mathbf{X}_h \times M_h$ such that

(2.22a) $$\langle A\mathbf{u}_h, \mathbf{w}_h\rangle - (p_h, \ \text{div}\ \mathbf{w}_h) = (\mathbf{f}, \mathbf{w}_h) \quad \forall \mathbf{w}_h \in \mathbf{X}_h$$

(2.22b) $$(\text{div}\ \mathbf{u}_h, q_h) = 0 \quad \forall q_h \in M_h \ .$$

The well-posedness of $(\mathscr{Q}_h) \equiv (\mathscr{R}_h)$ follows in analogous way to that of $(\mathscr{Q}) \equiv (\mathscr{R})$. In addition we have the discrete analogue of (2.14):

(2.23) $$\|\mathbf{u}_h\|_{D^r(\Omega)} \leq C(\|\mathbf{f}\|_{L^2(\Omega)}) \ ,$$

It follows from taking $\mathbf{w}_h \in \mathbf{V}_h \subset \mathbf{X}_h$ in (2.22a) that $\mathbf{u}_h$, the solution of $(\mathscr{Q}_h) \equiv (\mathscr{R}_h)$, is the unique $\mathbf{u}_h$ that solves $(\mathscr{S}_h)$. For the well-posedness of p_h in $(\mathscr{S}_h)$ we require a discrete version of (2.17a): There exists a constant $\beta_h > 0$ such that

(2.24a) $$\sup_{\mathbf{w}_h \in \mathbf{X}_h} (q_h, \ \text{div}\ \mathbf{w}_h)/\|\mathbf{w}_h\|_{\mathbf{X}} \geq \beta_h \|q_h\|_M \quad \forall q_h \in M_h \ ,$$

which from (2.18) and (2.29) is clearly equivalent to the discrete version of (2.17b): For any $\nu \in (1, \infty)$ there exists a constant $\beta_h(\nu) > 0$ such that with $\nu' = \nu/(\nu - 1)$

(2.24b) $$\sup_{\mathbf{w}_h \in \mathbf{X}_h} (q_h, \ \text{div}\ \mathbf{w}_h)/\|\mathbf{w}_h\|_{W^{1,\nu}(\Omega)} \geq \beta_h(\nu)\|q_h\|_{L^{\nu'}(\Omega)} \quad \forall q_h \in M_h \ .$$

It follows that $\beta_h \equiv \beta_h(r)$. If (2.24a) ($\equiv$ (2.24b)) holds then it is easily seen that there exists a unique solution to $(\mathscr{S}_h)$. However, in deriving sharp error bounds one needs to show that the constant $(\beta_h)^{-1}$ and possibly $(\beta_h(\nu))^{-1}$, for certain values of ν, are bounded above independently of h. Clearly $(\beta_h)^{-1}$ bounded above independently of h does not necessarily imply that $(\beta_h(\nu))^{-1}$ is and vice versa. Finally it is easily deduced, see p. 115 and Remark I.4.2 in G&R; that for any $\mathbf{v} \in \mathbf{V}$ and $\mathbf{w}_h \in \mathbf{X}_h$ there exists a $\mathbf{v}_h \in \mathbf{V}_h$ such that if (2.24a) holds then

(2.25a) $$\|\mathbf{v} - \mathbf{v}_h\|_{\mathbf{X}} \leq C((\beta_h)^{-1})\|\mathbf{v} - \mathbf{w_h}\|_{\mathbf{X}} \ ,$$

and if in addition $\mathbf{v} \in \mathbf{V} \cap [W^{1,\nu}(\Omega)]^2$ then

(2.25b) $$\|\mathbf{v} - \mathbf{v_h}\|_{W^{1,\nu}(\Omega)} \leq C((\beta_h(\nu))^{-1})\|\mathbf{v} - \mathbf{w_h}\|_{W^{1,\nu}(\Omega)} \ .$$

3. Abstract error bounds

We now prove an abstract error bound by applying Lemma 2.1 and Lemma 2.2. The key idea is to introduce a quasi-norm.

Let $(\mathbf{u}, p)$ be the unique solution of $(\mathscr{S})$. Then we define for any $\mathbf{w} \in [W^{1,r}(\Omega)]^2$

(3.1) $$|\mathbf{w}|^{\rho}_{(r,\alpha)} \equiv \int_{\Omega} |D(\mathbf{w})|^2[(|D(\mathbf{u})| + |D(\mathbf{w})|)^{\alpha}(1 + |D(\mathbf{u})| + |D(\mathbf{w})|)^{1-\alpha}]^{r-2} \ ,$$

where $\rho = \max\{r, 2\}$. Throughout this paper we will adopt the following notation: (i) if $s \in (1, \infty)$ then s' denotes its conjugate, i.e. $s' = s/(s-1)$, and (ii) for $1 < r \leq s \leq 2$ or $2 \leq s \leq r < \infty$ we set

$$\gamma \equiv \gamma(r,s) = \begin{cases} 2 & \text{if } r = 2 \\ s(2-r)/(2-s) & \text{if } r \neq 2 \text{ and } s \neq 2 \\ \infty & \text{if } r \neq 2 \text{ and } s = 2 \,. \end{cases} \tag{3.2}$$

It is then a simple matter to deduce using Hölder inequalities that

(a) for $1 < r \leq s \leq 2$ then if $\mathbf{u}, \mathbf{w} \in [W_0^{1,\gamma}(\Omega)]^2$, see B&L,

$$\|\mathbf{w}\|^2_{W^{1,s}(\Omega)} \leq C[1 + |\mathbf{u}|_{W^{1,\gamma}(\Omega)} + |\mathbf{w}|_{W^{1,\gamma}(\Omega)}]^{2-r} |\mathbf{w}|^2_{(r,\alpha)} \,; \tag{3.3a}$$

(b) for $r \in (1,2]$ and $\mathbf{w} \in [W^{1,\kappa}(\Omega)]^2$, $\kappa \in [r, 2 + \alpha(r-2)]$,

$$|\mathbf{w}|^2_{(r,\alpha)} \leq |\mathbf{w}|^\kappa_{D^\kappa(\Omega)} \leq C |\mathbf{w}|^\kappa_{W^{1,\kappa}(\Omega)} \,; \tag{3.3b}$$

(c) for $r \in [2,\infty)$, $\kappa \in [2 + \alpha(r-2), r]$ and $\mathbf{w} \in [W_0^{1,r}(\Omega)]^2$

$$C\|\mathbf{w}\|^\kappa_{W^{1,\kappa}(\Omega)} \leq |\mathbf{w}|^\kappa_{D^\kappa(\Omega)} \leq |\mathbf{w}|^r_{(r,\alpha)} \,; \tag{3.3c}$$

(d) for $2 \leq s \leq r < \infty$ then if $\mathbf{u}, \mathbf{w} \in [W^{1,\gamma}(\Omega)]^2$, see B&L,

$$|\mathbf{w}|^r_{(r,\alpha)} \leq C[1 + |\mathbf{u}|_{W^{1,\gamma}(\Omega)} + |\mathbf{w}|_{W^{1,\gamma}(\Omega)}]^{r-2} |\mathbf{w}|^2_{W^{1,s}(\Omega)} \,. \tag{3.3d}$$

From (3.3b&d) we see that $|\cdot|_{(r,\alpha)}$ is well-defined on $[W^{1,r}(\Omega)]^2$. Moreover, we have the following result.

Lemma 3.1. *$|\cdot|_{(r,\alpha)}$ is a quasi-norm on $\mathbf{X} \equiv [W_0^{1,r}(\Omega)]^2$; that is, it satisfies all the properties of a norm except the homogeneity property.*

Proof. Clearly $|\mathbf{w}|_{(r,\alpha)} \geq 0$ for all $\mathbf{w} \in \mathbf{X}$ and from (3.3a&c) we see that $|\mathbf{w}|_{(r,\alpha)} = 0$ only if $\mathbf{w} = 0$. In order to prove the triangle inequality, firstly we note for all $a \geq 0$ that the function $t \to t^2[(a+t)^\alpha(1+a+t)^{1-\alpha}]^{r-2}$ is monotonically increasing for $t \geq 0$. We further note that for all $a \geq 0$ and for all $\sigma_1 \geq \sigma_2 \geq 0$ that

$$\begin{aligned} &(\sigma_1+\sigma_2)^{2/\rho}[(a+\sigma_1+\sigma_2)^\alpha(1+a+\sigma_1+\sigma_2)^{1-\alpha}]^{(r-2)/\rho} \\ &\leq \sum_{i=1}^2 \sigma_i^{2/\rho}[a+\sigma_i)^\alpha(1+a+\sigma_i)^{1-\alpha}]^{(r-2)/\rho} \,. \end{aligned} \tag{3.4}$$

The inequality (3.4) is easily established if $r \in (1,2]$. If $r \in [2,\infty)$ then (3.4) clearly holds if $\sigma_2 = 0$. Therefore we can assume that $\sigma_2 > 0$ and the desired inequality then follows form noting, since $r \geq 2$, that

$$\begin{aligned} &\sum_{i=1}^2 \left(\frac{\sigma_i}{\sigma_1+\sigma_2}\right)^{2/r} \left[\left(\frac{a+\sigma_i}{a+\sigma_1+\sigma_2}\right)^\alpha \left(\frac{1+a+\sigma_i}{1+a+\sigma_1+\sigma_2}\right)^{1-\alpha}\right]^{(r-2)/r} \\ &\geq \sum_{i=1}^2 \left(\frac{\sigma_i}{\sigma_1+\sigma_2}\right) = 1 \,. \end{aligned}$$

Then for all $\mathbf{w}_1$, $\mathbf{w}_2 \in \mathbf{X}$ on setting

$$F(\mathbf{w}_i) \equiv |D(\mathbf{w}_i)|^2[(|D(\mathbf{u})| + |D(\mathbf{w}_i)|)^\alpha(1 + |D(\mathbf{u})| + |D(\mathbf{w}_i)|)^{1-\alpha}]^{r-2} \,,$$

we have, on applying a Hölder inequality, that

$$|\mathbf{w}_1+\mathbf{w}_2|^{\rho}_{(r,\alpha)} = \int_\Omega F(\mathbf{w}_1+\mathbf{w}_2) = \int_\Omega [F(\mathbf{w}_1+\mathbf{w}_2)]^{1/\rho}[F(\mathbf{w}_1+\mathbf{w}_2)]^{(\rho-1)/\rho}$$
$$\le \sum_{i=1}^{2}\int_\Omega [F(\mathbf{w}_i)]^{1/\rho}[F(\mathbf{w}_1+\mathbf{w}_2)]^{(\rho-1)/\rho}$$
$$\le \sum_{i=1}^{2}|\mathbf{w}_i|_{(r,\alpha)}|\mathbf{w}_1+\mathbf{w}_2|^{\rho-1}_{(r,\alpha)}$$

and hence the desired result. □

By introducing the quasi-norm $|\cdot|_{(r,\alpha)}$, we can now state our abstract error bounds in the following way.

Theorem 3.1. *Assume that $k(\cdot)$ satisfies Assumptions* (A). *Let $(\mathbf{u},p)\in\mathbf{V}\times M$ be the unique solution of $(\mathscr{S})$ and $\mathbf{u}_h\in\mathbf{V}_h$ be the unique solution of $(\mathscr{Q}_h)\equiv(\mathscr{R}_h)$. Then for all $\mathbf{v}_h\in\mathbf{V}_h$ and $q_h\in M_h$ we have under the required regularity on $(\mathbf{u},p)$:*

(i) *If $1<r\le s\le 2$ then*

$$|\mathbf{u}-\mathbf{u}_h|_{(r,\alpha)} \le C_1[|\mathbf{u}-\mathbf{v}_h|_{(r,\alpha)} + \|p-q_h\|_{L^{s'}(\Omega)}]\,, \tag{3.5a}$$

where $C_1\equiv C(\|\mathbf{u}\|_{W^{1,\gamma}(\Omega)},\|\mathbf{u}_h\|_{W^{1,\gamma}(\Omega)},\|\mathbf{v}_h\|_{W^{1,\gamma}(\Omega)})$.
(ii) *If $r\in[2,\infty)$ and $\kappa\in[2+\alpha(r-2),r]$ then*

$$|\mathbf{u}-\mathbf{u}_h|_{(r,\alpha)} \le C[|\mathbf{u}-\mathbf{v}_h|_{(r,\alpha)} + (\|p-q_h\|_{L^{\kappa'}(\Omega)})^{\kappa'/r}]\,. \tag{3.5b}$$

If in addition (2.24a) *holds, then $(\mathbf{u}_h,p_h)\in\mathbf{V}_h\times M_h$ the unique solution of $(\mathscr{S}_h)$ is such that*

(i) *If $r\in(1,2]$ and $\kappa\in[r,2+\alpha(r-2)]$, then*

$$\|p-p_h\|_{L^{\kappa'}(\Omega)} \le C((\beta_h(\kappa))^{-1})[(|\mathbf{u}-\mathbf{u}_h|_{(r,\alpha)})^{2/\kappa'} + \|p-q_h\|_{L^{\kappa'}(\Omega)}]\,. \tag{3.6a}$$

(ii) *If $2\le s\le r<\infty$ then*

$$\|p-p_h\|_{L^{s'}(\Omega)} \le C_2[(|\mathbf{u}-\mathbf{u}_h|_{(r,\alpha)})^{r/2} + \|p-q_h\|_{L^{s'}(\Omega)}]\,, \tag{3.6b}$$

where $C_2\equiv C((\beta_h(s))^{-1},\|\mathbf{u}\|_{W^{1,\gamma}(\Omega)},\|\mathbf{u}_h\|_{W^{1,\gamma}(\Omega)})$.

Proof. From $(\mathscr{S})$ and $(\mathscr{R}_h)$ it follows that

$$\langle A\mathbf{u}-A\mathbf{u}_h,\mathbf{v}_h\rangle = (p-q_h,\ \text{div}\ \mathbf{v}_h)\,. \tag{3.7}$$

Hence we have that

$$\langle A\mathbf{u}-A\mathbf{u}_h,\mathbf{u}-\mathbf{u}_h\rangle = \langle A\mathbf{u}-A\mathbf{u}_h,\mathbf{u}-\mathbf{v}_h\rangle + (p-q_h,\ \text{div}\ (\mathbf{v}_h-\mathbf{u}_h))\,. \tag{3.8}$$

Setting $\mathbf{e}=\mathbf{u}-\mathbf{u}_h$ and noting for all $X,Y\in\mathbb{M}_n$ that

$$(|X|+|Y|)/2 \le |X|+|X-Y| \le 2(|X|+|Y|)\,, \tag{3.9}$$

it follows from (3.8), (2.1) with $\delta = 0$ and (2.4) with ε sufficiently small that

$$\begin{aligned} M|\mathbf{e}|^{\rho}_{(r,\alpha)} &\leq \langle A\mathbf{u} - A\mathbf{u}_h, \mathbf{u} - \mathbf{u}_h \rangle \\ &\leq C \int_\Omega [(|D(\mathbf{u})| + |D(\mathbf{e})|)^\alpha (1 + |D(\mathbf{u})| + |D(\mathbf{e})|)^{1-\alpha}]^{r-2} |D(\mathbf{e})| |D(\mathbf{u} - \mathbf{v}_h)| \\ &\quad + |(p - q_h,\ \text{div}\,(\mathbf{v}_h - \mathbf{u}_h))| \\ &\leq C[|\mathbf{u} - \mathbf{v}_h|^{\rho}_{(r,\alpha)} + |(p - q_h,\ \text{div}\,(\mathbf{v}_h - \mathbf{u}_h))|] \,. \end{aligned} \tag{3.10}$$

Next if $1 < r \leq s \leq 2$, then we note from (3.3a) that for all $\varepsilon > 0$

$$\begin{aligned} |(p - q_h, \text{div}\,(\mathbf{v}_h - \mathbf{u}_h))| &\leq \varepsilon[\|\mathbf{e}\|^2_{W^{1,s}(\Omega)} + \|\mathbf{u} - \mathbf{v}_h\|^2_{W^{1,s}(\Omega)}] \\ &\quad + C\varepsilon^{-1}\|p - q_h\|^2_{L^{s'}(\Omega)} \leq \varepsilon C_1[|\mathbf{e}|^2_{(r,\alpha)} + |\mathbf{u} - \mathbf{v}_h|^2_{(r,\alpha)}] + C\varepsilon^{-1}\|p - q_h\|^2_{L^{s'}(\Omega)} \,. \end{aligned} \tag{3.11a}$$

Alternatively if $r \in [2, \infty)$, then we note from (3.3c) and a Young's inequality that for all $\varepsilon > 0$

$$\begin{aligned} |(p - q_h, \text{div}\,(\mathbf{v}_h - \mathbf{u}_h))| &\leq \varepsilon[\|\mathbf{e}\|^\kappa_{W^{1,\kappa}(\Omega)} + \|\mathbf{u} - \mathbf{v}_h\|^\kappa_{W^{1,\kappa}(\Omega)}] \\ &\quad + C(\varepsilon^{-1})\|p - q_h\|^{\kappa'}_{L^{\kappa'}(\Omega)} \leq \varepsilon C[|\mathbf{e}|^r_{(r,\alpha)} + |\mathbf{u} - \mathbf{v}_h|^r_{(r,\alpha)}] + C(\varepsilon^{-1})\|p - q_h\|^{\kappa'}_{L^{\kappa'}(\Omega)} \,. \end{aligned} \tag{3.11b}$$

Combining (3.10) with (3.11a&b), with ε sufficiently small, yields the desired results (3.5a&b).

If (2.24a) holds, then from ($\mathscr{S}$) and ($\mathscr{S}_h$) it follows that

$$(q_h - p_h, \text{div}\,\mathbf{w}_h) = \langle A\mathbf{u} - A\mathbf{u}_h, \mathbf{w}_h \rangle + (q_h - p, \text{div}\,\mathbf{w}_h)$$

and hence for $\nu \in (1, \infty)$

$$\begin{aligned} &\beta_h(\nu)\|q_h - p_h\|_{L^{\nu'}(\Omega)} \\ &\quad \leq \sup_{\mathbf{w}_h \in \mathbf{X}_h} |\langle A\mathbf{u} - A\mathbf{u}_h, \mathbf{w}_h \rangle / \|\mathbf{w}_h\|_{W^{1,\nu}(\Omega)}| + \|p - q_h\|_{L^{\nu'}(\Omega)} \,. \end{aligned} \tag{3.12a}$$

Next we note from (2.1a) with $\delta = 0$ and (3.9) that

$$\begin{aligned} &\sup_{\mathbf{w}_h \in \mathbf{X}_h} |\langle A\mathbf{u} - A\mathbf{u}_h, \mathbf{w}_h \rangle / \|\mathbf{w}_h\|_{W^{1,\nu}(\Omega)}| \\ &\quad \leq \|k(|D(\mathbf{u})|)D(\mathbf{u}) - k(|D(\mathbf{u}_h)|)D(\mathbf{u}_h)\|_{L^{\nu'}(\Omega)} \\ &\quad \leq C\|[(|D(\mathbf{u})| + |D(\mathbf{e})|)^\alpha (1 + |D(\mathbf{u})| + |D(\mathbf{e})|)^{1-\alpha}]^{(r-2)} |D(\mathbf{e})|\|_{L^{\nu'}(\Omega)} \,. \end{aligned} \tag{3.12b}$$

If $r \in (1, 2]$ and $\kappa = 2 + \mu(r - 2)$ for a $\mu \in [\alpha, 1]$, then choosing $\nu = \kappa$ in (3.12) and noting that

$$\begin{aligned} &\|[(|D(\mathbf{u})| + |D(\mathbf{e})|)^\alpha (1 + |D(\mathbf{u})| + |D(\mathbf{e})|)^{1-\alpha}]^{(r-2)} |D(\mathbf{e})|\|_{L^{\kappa'}(\Omega)} \\ &\quad \leq C_3[|\mathbf{e}|_{(r,\alpha)}]^{2/\kappa'} \,, \end{aligned}$$

where $C_3 \equiv C(\|[(|D(\mathbf{u})| + |D(\mathbf{e})|)^\alpha (1 + |D(\mathbf{u})| + |D(\mathbf{e})|)^{1-\alpha}]^{(r-2)} |D(\mathbf{e})|^{(2-\kappa)}\|_{L^\infty(\Omega)}) \leq C$, yields the desired result (3.6a). If $2 \leq s \leq r < \infty$, then choosing $\nu = s$ in (3.12) and noting from a Hölder inequality that

$$\|[(|D(\mathbf{u})| + |D(\mathbf{e})|)^{\alpha}(1 + |D(\mathbf{u})| + |D(\mathbf{e})|)^{1-\alpha}]^{(r-2)}|D(\mathbf{e})|\|_{L^{s'}(\Omega)}$$
$$\leq C(\|\mathbf{u}\|_{W^{1,\gamma}(\Omega)}, \|\mathbf{u}_h\|_{W^{1,\gamma}(\Omega)})|\mathbf{e}|^{r/2}_{(r,\alpha)} ,$$

yields the desired result (3.6b). □

We believe the introduction of the quasi-norm leads to a very clean and simple abstract theory for a general viscosity law; compare the bounds (3.5) and (3.6) with those in Sandri (1993) for example. From (3.5) and (3.6) one can recover all the known error bounds in the literature, see Sect. 1, and in many cases improve on them. By exploring the relationships (3.3) between the quasi-norm and the standard Sobolev norms we obtain the following abstract error bounds.

Theorem 3.2. *Assume that $k(\cdot)$ satisfies Assumptions* (A). *Let $(\mathbf{u}, p) \in \mathbf{V} \times M$ be the unique solution of $(\mathscr{S})$. Then for all $\mathbf{v}_h \in \mathbf{V}_h$, $q_h \in M_h$ and $\mathbf{w}_h \in \mathbf{X}_h$ we have under the required regularity on $(\mathbf{u}, p)$:*

(i) *If $1 < r \leq s \leq 2$ and $\kappa \in [r, 2 + \alpha(r-2)]$, then $\mathbf{u}_h \in \mathbf{V}_h$, the unique solution of $(\mathscr{Q}_h) \equiv (\mathscr{R}_h)$, is such that*

$$|\mathbf{u} - \mathbf{u}_h|_{(r,\alpha)} \leq C_1[(\|\mathbf{u} - \mathbf{v}_h\|_{W^{1,\kappa}(\Omega)})^{\kappa/2} + \|p - q_h\|_{L^{s'}(\Omega)}] \tag{3.13a}$$

and consequently

$$\|\mathbf{u} - \mathbf{u}_h\|_{W^{1,s}(\Omega)} \leq C_1[(\|\mathbf{u} - \mathbf{v}_h\|_{W^{1,\kappa}(\Omega)})^{\kappa/2} + \|p - q_h\|_{L^{s'}(\Omega)}] \tag{3.13b}$$

where $C_1 \equiv C(\|\mathbf{u}\|_{W^{1,\gamma}(\Omega)}, \|\mathbf{u}_h\|_{W^{1,\gamma}(\Omega)}, \|\mathbf{v}_h\|_{W^{1,\gamma}(\Omega)})$.

If in addition to the above (2.24a) *holds, then $(\mathbf{u}_h, p_h) \in \mathbf{V}_h \times M_h$ the unique solution of $(\mathscr{S}_h)$ is such that*

$$\|\mathbf{u} - \mathbf{u}_h\|_{W^{1,s}(\Omega)} \leq C_2[(\|\mathbf{u} - \mathbf{w}_h\|_{W^{1,\kappa}(\Omega)})^{\kappa/2} + \|p - q_h\|_{L^{s'}(\Omega)}] , \tag{3.13c}$$

$$\|p - p_h\|_{L^{\kappa'}(\Omega)} \leq C_2\{[\|\mathbf{u} - \mathbf{w}_h\|_{W^{1,\kappa}(\Omega)} + (\|p - q_h\|_{L^{s'}(\Omega)})^{2/\kappa}]^{\kappa-1} + \|p - q_h\|_{L^{\kappa'}(\Omega)}\} , \tag{3.13d}$$

where $C_2 \equiv C(\|\mathbf{u}\|_{W^{1,\gamma}(\Omega)}, \|\mathbf{u}_h\|_{W^{1,\gamma}(\Omega)}, \|\mathbf{w}_h\|_{W^{1,\gamma}(\Omega)}, (\beta_h(\kappa))^{-1}, (\beta_h(\gamma))^{-1})$.

(ii) *If $r \in [2, \infty)$ and $\kappa \in [2 + \alpha(r-2), r]$, then $\mathbf{u}_h \in \mathbf{V}_h$, the unique solution of $(\mathscr{Q}_h) \equiv (\mathscr{R}_h)$, is such that*

$$|\mathbf{u} - \mathbf{u}_h|_{(r,\alpha)} \leq C_3[\|\mathbf{u} - \mathbf{v}_h\|^2_{W^{1,2}(\Omega)} + \|p - q_h\|^{\kappa'}_{L^{\kappa'}(\Omega)}]^{1/r} , \tag{3.14a}$$

and consequently

$$\|\mathbf{u} - \mathbf{u}_h\|_{W^{1,\kappa}(\Omega)} \leq C_3[\|\mathbf{u} - \mathbf{v}_h\|^2_{W^{1,2}(\Omega)} + \|p - q_h\|^{\kappa'}_{L^{\kappa'}(\Omega)}]^{1/\kappa} , \tag{3.14b}$$

where $C_3 \equiv C(\|\mathbf{u}\|_{W^{1,\infty}(\Omega)}, \|\mathbf{v}_h\|_{W^{1,\infty}(\Omega)})$.

If in addition to the above (2.24a) *holds, then $(\mathbf{u}_h, p_h) \in \mathbf{V}_h \times M_h$ the unique solution of $(\mathscr{S}_h)$ is such that*

$$\|\mathbf{u} - \mathbf{u}_h\|_{W^{1,\kappa}(\Omega)} \leq C_4[\|\mathbf{u} - \mathbf{w}_h\|^2_{W^{1,2}(\Omega)} + \|p - q_h\|^{\kappa'}_{L^{\kappa'}(\Omega)}]^{1/\kappa} , \tag{3.14c}$$

where $C_4 \equiv C(\|\mathbf{u}\|_{W^{1,\infty}(\Omega)}, \|\mathbf{w}_h\|_{W^{1,\infty}(\Omega)}, (\beta_h(2))^{-1}, (\beta_h(\infty))^{-1})$*; and if* $2 \leq s \leq r < \infty$ *then*

$$\|p - p_h\|_{L^{s'}(\Omega)} \leq C_5 \left\{ [\|\mathbf{u} - \mathbf{w}_h\|^2_{W^{1,2}(\Omega)} + \|p - q_h\|^{\kappa'}_{L^{\kappa'}(\Omega)}]^{1/2} + \|p - q_h\|_{L^{s'}(\Omega)} \right\} , \tag{3.14d}$$

where $C_5 \equiv (C_4, (\beta_h(s))^{-1}, \|\mathbf{u}_h\|_{W^{1,\gamma}(\Omega)})$.

Proof. (i) The desired results (3.13a&b) follow immediately from (3.5a) and (3.3a&b). (3.13c&d) follow from (3.13a&b), (3.6a) and (2.25b).

(ii) The results (3.14a&b) follow immediately from (3.5b) and (3.3c&d) with $s = 2$. (3.14c&d) follow from (3.14a&b), (3.6b) and (2.25b). □

Remark 3.1. Due to the limited regularity of the solution $(\mathbf{u}, p)$ of $(\mathscr{S})$ for degenerate $k(\cdot)$, i.e. $\alpha \in (0, 1]$; it is important to prove error bounds for the approximation $(\mathscr{S}_h)$ under the weakest regularity requirement. Hence the choice of $s = 2$ in (3.3d) in proving the results (3.14) above.

4. Error bounds for some finite element approximations

In this section, we consider some finite element approximations of $(\mathscr{S})$, which satisfy the abstract theory of the previous sections. We assume throughout, for ease of exposition, that Ω is a convex polygonal domain. Let T^h be a partitioning of Ω into disjoint open regular triangles τ, each of maximum diameter bounded above h, so that $\overline{\Omega} \equiv \cup_{\tau \in T^h} \overline{\tau}$. Consider the following assumptions for an integer $m \geq 1$ and any $\nu \in [1, \infty]$.

(H1): Approximation property of $\mathbf{X}_h$.
There is a continuous linear operator $\pi_h : [W^{1,\nu}_0(\Omega)]^2 \to \mathbf{X}_h$ such that for $j = 0 \to m$

$$\|\mathbf{w} - \pi_h \mathbf{w}\|_{W^{1,\nu}(\Omega)} \leq C h^j \|\mathbf{w}\|_{W^{j+1,\nu}(\Omega)} \quad \forall \mathbf{w} \in [W^{j+1,\nu}(\Omega) \cap W^{1,\nu}_0(\Omega)]^2 .$$

(H2): Approximation property of M_h.
There is a continuous linear operator $\rho_h : L^\nu(\Omega) \to M_h$ such that for all $j = 0 \to m$

$$\|q - \rho_h q\|_{L^\nu(\Omega)} \leq C h^j \|q\|_{W^{j,\nu}(\Omega)} \quad \forall q \in W^{j,\nu}(\Omega) .$$

(H3): Inverse property of $\mathbf{X}_h$.
For $\nu > \mu \geq 1$

$$\|\mathbf{w}_h\|_{W^{1,\nu}(\Omega)} \leq C h^{2(\nu^{-1} - \mu^{-1})} \|\mathbf{w}_h\|_{W^{1,\mu}(\Omega)} \quad \forall \mathbf{w}_h \in \mathbf{X}_h .$$

Theorem 4.1. *Assume that* $k(\cdot)$ *satisfies Assumptions* (A). *Let* $(\mathbf{u}, p) \in \mathbf{V} \times M$ *be the unique solution of* $(\mathscr{S})$. *Then if* (H1) *and* (H2) *hold for* $j = 0 \to m$ *we have under the required regularity on* $(\mathbf{u}, p)$*:*

(i) *If* $r \in (1, 2]$, $\kappa \in [r, 2 + \alpha(r - 2)]$ *and if* (2.24a) *holds, then* $(\mathbf{u}_h, p_h) \in \mathbf{V}_h \times M_h$ *the unique solution of* $(\mathscr{S}_h)$ *is such that*

$$\|\mathbf{u} - \mathbf{u}_h\|_{W^{1,r}(\Omega)} + (\|p - p_h\|_{L^{\kappa'}(\Omega)})^{\kappa/[2(\kappa-1)]} \leq C_1 h^{\kappa j/2} , \tag{4.1}$$

where $C_1 \equiv C(\|\mathbf{u}\|_{W^{j+1,\kappa}(\Omega)}, \|p\|_{W^{j,r'}(\Omega)}, (\beta_h)^{-1}, (\beta_h(\kappa))^{-1})$.

(ii) *If* $r \in [2,\infty)$, $\kappa \in [2+\alpha(r-2), r]$ *and if* (2.24a) *holds, then* $(\mathbf{u}_h, p_h) \in \mathbf{V}_h \times M_h$ *the unique solution of* $(\mathscr{S}_h)$ *is such that*

$$\|\mathbf{u} - \mathbf{u}_h\|_{W^{1,\kappa}(\Omega)} \le C_2 h^{j/(\kappa-1)}, \tag{4.2a}$$

where $C_2 \equiv C(\|\mathbf{u}\|_{W^{1,\infty}(\Omega)}, \|\mathbf{u}\|_{W^{j+1,2}(\Omega)}, \|p\|_{W^{j,\kappa'}(\Omega)}, (\beta_h(2))^{-1}, (\beta_h(\infty))^{-1})$*; and*

$$\|p - p_h\|_{L^{r'}(\Omega)} \le C(C_2, (\beta_h)^{-1}) h^{j\kappa/[2(\kappa-1)]}. \tag{4.2b}$$

Proof. The results (4.1) and (4.2) follow immediately from (3.13c&d) and (3.14c&d), with $\mathbf{w}_h \equiv \pi_h \mathbf{u}$, $q_h \equiv \rho_h p$ and $s = r$, and (2.23). □

If $k(\cdot)$ is non-degenerate; that is, $\alpha = 0$ – see Remark 2.1 for examples; then (4.1) and (4.2) yield optimal error bounds for the velocity and pressure on choosing $\kappa = 2$ and assuming the required regularity. We note that these error bounds deteriorate as $k(\cdot)$ becomes more degenerate, i.e. as $\alpha \to 1$. Below we show that under assumption (H3), the above error bounds for the velocity if $r \in (1,2)$ and for the pressure if $r \in (2,\infty)$ hold in stronger norms.

Theorem 4.2. *Assume that* $k(\cdot)$ *satisfies Assumptions* (A). *Let* $(\mathbf{u}, p) \in \mathbf{V} \times M$ *be the unique solution of* $(\mathscr{S})$. *Then if* (H1)→(H3) *hold for* $j = 1 \to m$ *we have under the required regularity on* $(\mathbf{u}, p)$*:*

(i) *If* $r \in (1,2]$, $\kappa \in [r, 2+\alpha(r-2)]$ *and if* (2.24a) *holds, then* $(\mathbf{u}_h, p_h) \in \mathbf{V}_h \times M_h$ *the unique solution of* $(\mathscr{S}_h)$ *is such that*

$$\|\mathbf{u} - \mathbf{u}_h\|_{W^{1,s}(\Omega)} \le C_1 h^{\kappa j/2}, \tag{4.3}$$

where $s = 4r/[4-(2-r)\kappa] \in [\kappa, 2]$ *if* $j = 1$ *and* $s = 2$ *otherwise, and* $C_1 \equiv C(\|\mathbf{u}\|_{W^{1,\infty}(\Omega)}, \|\mathbf{u}\|_{W^{j+1,s}(\Omega)}, \|p\|_{W^{j,r'}(\Omega)}, (\beta_h)^{-1}, (\beta_h(\kappa))^{-1})$.

(ii) *If* $r \in [2,\infty)$, $\kappa \in [2+\alpha(r-2), r]$ *and if* (2.24a) *holds, then* $(\mathbf{u}_h, p_h) \in \mathbf{V}_h \times M_h$ *the unique solution of* $(\mathscr{S}_h)$ *is such that*

$$\|p - p_h\|_{L^{s'}(\Omega)} \le C_2 h^{j\kappa/[2(\kappa-1)]}, \tag{4.4}$$

where $s' = \max\{4\kappa(\kappa-1)/[2\kappa(\kappa-1)+(r-2)(\kappa-2)], r'\}$ *if* $j = 1$ *and* $s' = 2$ *if* $j \ge 2$, $C_2 \equiv C(\|\mathbf{u}\|_{W^{1,\infty}(\Omega)}, \|\mathbf{u}\|_{W^{j+1,2}(\Omega)}, \|p\|_{W^{j,\sigma'}(\Omega)}, (\beta_h(2))^{-1}, (\beta_h(\infty))^{-1}, (\beta_h(s))^{-1})$ *and* $\sigma' = \max\{s', \kappa'\}$.

Proof. (i) Let $\xi = j\kappa$. Firstly, from (H3) we have for $\gamma(i) \ge s(i) \ge 1$ that

$$\begin{aligned}
&\|\mathbf{u}_h\|_{W^{1,\gamma(i)}(\Omega)} \\
&\le \|\mathbf{u}_h - \pi_h\mathbf{u}\|_{W^{1,\gamma(i)}(\Omega)} + \|\pi_h\mathbf{u}\|_{W^{1,\gamma(i)}(\Omega)} \\
&\le Ch^{2([\gamma(i)]^{-1}-[s(i)]^{-1})}\|\mathbf{u}_h - \pi_h\mathbf{u}\|_{W^{1,s(i)}(\Omega)} + \|\pi_h\mathbf{u}\|_{W^{1,\gamma(i)}(\Omega)} \\
&\le Ch^{2([\gamma(i)]^{-1}-[s(i)]^{-1})}[\|\mathbf{u} - \mathbf{u}_h\|_{W^{1,s(i)}(\Omega)} + \|\mathbf{u} - \pi_h\mathbf{u}\|_{W^{1,s(i)}(\Omega)}] \\
&\qquad + \|\mathbf{u}\|_{W^{1,\gamma(i)}(\Omega)}.
\end{aligned} \tag{4.5a}$$

Hence noting (H3) we have that if (4.3) holds for $s = s(i)$, i.e. $\|\mathbf{u} - \mathbf{u}_h\|_{W^{1,s(i)}(\Omega)} \leq C_1 h^{\xi/2}$ then

$$\|\mathbf{u}_h\|_{W^{1,\gamma(i)}(\Omega)} \leq C_1[1 + h^{2([\gamma(i)]^{-1} - [s(i)]^{-1})} h^{\xi/2}] \leq C_1 \,, \tag{4.5b}$$

where $\gamma(i) = \infty$ if $s(i) \geq 4/\xi$ and $\gamma(i) = 4s(i)/[4 - \xi s(i)]$ otherwise. Noting the inverse relation of (3.2), $s = s(r, \gamma) = 2\gamma/(2 - r + \gamma)$, (H1) and (H2) we have that (3.13c), with $s = s(i+1) \equiv 2\gamma(i)/[2 - r + \gamma(i)]$ and $\gamma = \gamma(i)$, yields the result (4.3) with $s = s(i+1)$. Therefore noting the above and (4.1) we have $\|\mathbf{u} - \mathbf{u}_h\|_{W^{1,s(i)}(\Omega)} \leq C_1 h^{\xi/2}$ for $i \geq 0$, where $s(0) = r$ and for $i \geq 0$

$$\begin{aligned} &s(i+1) = 2 \text{ if } s(i) \geq 4/\xi \\ &s(i+1) = F(s(i)) \equiv 8s(i)/[(2 - r)(4 - \xi s(i)) + 4s(i)] \text{ otherwise} \,. \end{aligned} \tag{4.6}$$

We now analyse this map.

It follows from (4.1) that the result (4.3) holds with $s = r$. Therefore we need only consider the case $r \in (1, 2)$ and then we have:

If $\xi \geq 4/r$, i.e. $j \geq 4\kappa/r \geq 4$; then (4.6) yields that $s(1) = 2$.

If $\xi \in (2, 4/r)$, i.e. $j \geq 2$, then it is easily established that $F : [r, 4/\xi] \to [a, 2]$, where $a \in (r, 2)$, and $F(x) > x$ for all $x \in [r, 4/\xi]$. Therefore it is impossible for $s(i) \in [r, 4/\xi)$ for all $i \geq 0$, and so $s(i) = 2$ for some i.

If $\xi \leq 2$, i.e. $j = 1$; then it is easily established that $F : [r, s_*] \to [a, s_*]$, where $s_* = 4r/[4 - (2 - r)\xi] \in (r, 2]$ and $a \in (r, s_*)$, $F(x) > x$ for all $x \in [r, s_*)$ and $F(s_*) = s_*$.

Hence the desired result (4.3).

(ii) From (4.5a) with $s(0) = \kappa$, (H3), (4.2a) and (2.23) we have that $\|\mathbf{u}_h\|_{W^{1,\gamma(0)}(\Omega)} \leq C_2$, where $\gamma(0) = \infty$ if $j \geq 2$ or $\kappa = 2$ and $\gamma(0) = \max\{r, 2\kappa(\kappa - 1)/(\kappa - 2)\}$ otherwise. Noting the inverse relation of (3.2), $s = s(r, \gamma) = 2\gamma/(2 - r + \gamma)$, (H1) and (H2) we have that (3.14d) with $s = s(r, \gamma(0)) = 2\gamma(0)/(2 - r + \gamma(0))$ and $\gamma = \gamma(0)$ yields the desired result (4.4). □

The assumptions (H1) and (H2) are standard and are easily satisfied. For example choose $\mathbf{X}_h$ and M_h to be continuous piecewise polynomials of degree m and $m - 1$, respectively, satisfying the required constraints. The inverse assumption (H3) also holds if the triangulation is quasi-uniform. However, choosing $\mathbf{X}_h$ and M_h so that $(\beta_h(\nu))^{-1}$, for the various required values of ν, is bounded above independently of h is much more delicate. The case $\nu = 2$ has been widely studied as this condition is required in approximating the linear Stokes problem, $r = 2$; see G&R for a full account of this problem. Below we study in detail a particular choice of spaces $\mathbf{X}_h$ and M_h, introduced by Fortin (1981) and Bernardi and Raugel (1985) for the Stokes problem, for which all the above assumptions can be checked explicitly.

Let P_m be the space of all polynomials of degree less than or equal to m. Let τ be an arbitrary triangle of T^h with vertices a_1, a_2 and a_3. We denote by e_i the edge opposite a_i and by $\mathbf{n}_i$ and $\mathbf{t}_i$ the unit outward normal and unit tangent to e_i. Let

$$\mathit{\Lambda}_1(\tau) \equiv \{[P_1]^2 + \text{ span } (\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3)\} \,, \tag{4.7}$$

where $\mathbf{p}_1 \equiv \mathbf{n}_1\lambda_2\lambda_3$, $\mathbf{p}_2 \equiv \mathbf{n}_2\lambda_3\lambda_1$, $\mathbf{p}_3 \equiv \mathbf{n}_3\lambda_1\lambda_2$, $\lambda_i \in P_1$ and $\lambda_i(a_j) = \delta_{ij}$ $i, j = 1 \to 3$. We then set

$$\mathbf{X}_h \equiv \{\mathbf{w}_h \in [C(\overline{\Omega})]^2 : \mathbf{w}_h|_\tau \in \mathit{\Lambda}_1(\tau) \quad \forall \tau \in T^h \text{ and } \mathbf{w}_h = 0 \text{ on } \mathit{\Gamma}\} \tag{4.8a}$$

and

$$M_h \equiv \{q_h \in L_0^{r'}(\Omega) : q_h|_\tau \in P_0 \quad \forall \tau \in T^h\} \,, \tag{4.8b}$$

noting that they satisfy (2.18). For $\nu \in [1,\infty]$ let $\pi_h : [W_0^{1,\nu}(\Omega)]^2 \to \mathbf{X}_h$ be such that

$$\pi_h \mathbf{w}(a) = R_h \mathbf{w}(a) \quad \text{for all vertices } a \text{ of } T^h \tag{4.9a}$$

and

$$\int_e (\mathbf{w} - \pi_h \mathbf{w}) \cdot \mathbf{n} = 0 \quad \text{for all edges } e \text{ of } T^h \,, \tag{4.9b}$$

where R_h is the (generalised, if $\nu \le 2$, as introduced in Clément (1975), see also Sect. A.3 in G&R) continues piecewise linear interpolant. It follows directly from (4.9b) that for all $\tau \in T^h$

$$\int_\tau \text{div } (\mathbf{w} - \pi_h \mathbf{w}) = 0 \quad \forall \mathbf{w} \in [W_0^{1,\nu}(\Omega)]^2 \tag{4.10}$$

and hence $\pi_h \mathbf{u} \in \mathbf{V}_h$, where $\mathbf{u}$ is the unique soluiton of $\mathscr{Q}) \equiv (\mathscr{R})$.

As the triangulation is regular, it is a simple matter to extend the results in Sect II.2.1 of G&R to show that:

(a) The following refined version of (H1) with $m = 1$ holds. For $j = 0 \to 1$ and for all $\tau \in T^h$ we have that

$$\begin{aligned} &\|\mathbf{w} - \pi_h \mathbf{w}\|_{W^{1,\nu}(\tau)} \le C h^j |\mathbf{w}|_{W^{j+1,\nu}(\Delta_\tau)} \\ &\qquad \forall \mathbf{w} \in [W^{j+1,\nu}(\Delta_\tau) \cap W_0^{1,\nu}(\Omega)]^2 \,, \end{aligned} \tag{4.11}$$

where for $\nu \in [1,2]$ Δ_τ denotes the union of all elements of T^h that share at least a vertex with τ and for $\nu \in (2,\infty]$ $\Delta_\tau \equiv \tau$.

(b) (H2) with $m = 1$ holds.

(c) (H3) holds if the triangulation is quasi-uniform.

It follows also from (4.10), (4.11) with $j = 0$, (2.18) and (2.17b) that

(d) (2.24b) holds for all $\nu \in (1,\infty)$ with $\beta_h(\nu) \ge C\beta(\nu)$.

Theorem 4.3. *Assume that $k(\cdot)$ satisfies Assumptions* (A). *Let $(\mathbf{u}, p) \in \mathbf{V} \times M$ be the unique solution to $(\mathscr{S})$. Let $\mathbf{X}_h$ and M_h be given as in* (4.8), *then there exists a unique solution $(\mathbf{u}_h, p_h) \in \mathbf{V}_h \times M_h$ to $(\mathscr{S}_h)$ and it follows under the required regularity on $(\mathbf{u}, p)$ that*

(i) *If $r \in (1,2]$ and $\kappa \in [r, 2 + \alpha(r-2)]$ then*

$$\begin{aligned} &\|\mathbf{u} - \mathbf{u}_h\|_{W^{1,r}(\Omega)} + (\|p - p_h\|_{L^{\kappa'}(\Omega)})^{\kappa/[2(\kappa-1)]} \\ &\qquad \le C(\|\mathbf{u}\|_{W^{2,\kappa}(\Omega)} \,, \|p\|_{W^{1,r'}(\Omega)}) h^{\kappa/2} \,. \end{aligned} \tag{4.12a}$$

Furthermore if the mesh is quasi-uniform, we have for $s = 4r/[4 - (2-r)\kappa] \in [\kappa, 2]$ that

$$\|\mathbf{u} - \mathbf{u}_h\|_{W^{1,s}(\Omega)} \le C(\|\mathbf{u}\|_{W^{1,\infty}(\Omega)} \,, \|\mathbf{u}\|_{W^{2,s}(\Omega)}, \ \|p\|_{W^{1,r'}(\Omega)}) h^{\kappa/2} \,. \tag{4.12b}$$

(ii) *If* $r \in [2,\infty)$ *and* $\kappa \in [2+\alpha(r-2), r]$ *then*

$$\begin{aligned} &\|\mathbf{u}-\mathbf{u}_h\|_{W^{1,\kappa}(\Omega)} + (\|p-p_h\|_{L^{r'}(\Omega)})^{2/\kappa} \\ &\quad \le C(\|\mathbf{u}\|_{W^{1,\infty}(\Omega)}, \|\mathbf{u}\|_{W^{2,2}(\Omega)}, \|p\|_{W^{1,\kappa'}(\Omega)}) h^{1/(\kappa-1)} . \end{aligned} \tag{4.13a}$$

Furthermore if the mesh is quasi-uniform and $2\kappa(\kappa-1) \ge (\kappa-2)r$ *we have for* $s' = 4\kappa(\kappa-1)/[2\kappa(\kappa-1)+(r-2)(\kappa-2)]$ *that*

$$\|p-p_h\|_{L^{s'}(\Omega)} \le C(\|\mathbf{u}\|_{W^{1,\infty}(\Omega)}, \|\mathbf{u}\|_{W^{2,2}(\Omega)}, \|p\|_{W^{1,\sigma'}(\Omega)}) h^{\kappa/[2(\kappa-1)]} , \tag{4.13b}$$

where $\sigma' = \max\{s', \kappa'\}$.

Proof. (i) From (d) above and (2.24b) it follows that there exists a unique solution $(\mathbf{u}_h, p_h) \in \mathbf{V}_h \times M_h$ to $(\mathscr{S}_h)$. The results (4.12a&b) follow directly from (4.1) and (4.3).

(ii) The result (4.13a) follows directly form (3.14a&b) and (3.6b) with $\mathbf{v}_h = \pi_h \mathbf{u} \in \mathbf{V}_h$ and $q_h = \rho_h p$, and noting (a), (b) and (d). The result (4.13b) follows similarly, after noting that (4.13a) and (c) yield that $\|\mathbf{u}_h\|_{W^{1,\gamma}(\Omega)} \le C$, see the proof of Theorem 4.2(ii). □

Finally for the case $r \in (1,2]$ and for this Fortin element, by further exploring the structure of the quasi-norm, we prove an optimal error bound for the velocity even for a degenerate $k(\cdot)$ under a strong regularity requirement on $\mathbf{u}$. The following generalises and improves on Theorem 4.4 in B&L.

Theorem 4.4. *Assume that* $k(\cdot)$ *satisfies Assumptions* (A). *Let* $(\mathbf{u}, p) \in V \times M$ *be the unique solution to* $(\mathscr{S})$. *Let* $\mathbf{X}_h$ *and* M_h *be given as in (4.8), then there exists a unique solution* $(\mathbf{u}_h, p_h) \in \mathbf{V}_h \times M_h$ *to* $(\mathscr{S}_h)$ *and it follows under the required regularity on* $(\mathbf{u}, p)$ *that if* $r \in (1,2]$ *and* $\kappa = 2+\alpha(r-2)$ *then*

$$\begin{aligned} &\|\mathbf{u}-\mathbf{u}_h\|_{W^{1,r}(\Omega)} + \left(\|p-p_h\|_{L^{\kappa'}(\Omega)}\right)^{\kappa/[2(\kappa-1)]} \\ &\quad \le C\left(\|\mathbf{u}\|_{C^{2,(2-\kappa)/\kappa}(\overline{\Omega})}, \|\mathbf{u}\|_{W^{3,1}(\Omega)}, \|p\|_{W^{1,r'}(\Omega)}\right) h . \end{aligned} \tag{4.14a}$$

Furthermore if the mesh is quasi-uniform, we have

$$\begin{aligned} &\|\mathbf{u}-\mathbf{u}_h\|_{W^{1,2}(\Omega)} \\ &\quad \le C\left(\|\mathbf{u}\|_{C^{2,(2-\kappa)/\kappa}(\overline{\Omega})}, \|\mathbf{u}\|_{W^{3,1}(\Omega)}, \|p\|_{W^{1,r'}(\Omega)}\right) h . \end{aligned} \tag{4.14b}$$

Proof. From (d) above and (2.24b) it follows that there exists a unique solution $(\mathbf{u}_h, p_h) \in \mathbf{V}_h \times M_h$ to $(\mathscr{S}_h)$. Choosing $\mathbf{v}_h = \pi_h \mathbf{u}$ and $q_h = \rho_h p$ in (3.5a) and noting (3.3a) yields for $s \in [r,2]$

$$\begin{aligned} \|\mathbf{u}-\mathbf{u}_h\|_{W^{1,s}(\Omega)} &\le C_1 |\mathbf{u}-\mathbf{u}_h|_{(r,\alpha)} \\ &\le C_1 \left[|\mathbf{u}-\pi_h\mathbf{u}|_{(r,\alpha)} + \|p-\rho_h p\|_{L^{s'}(\Omega)}\right] , \end{aligned} \tag{4.15}$$

where $C_1 \equiv C(\|\mathbf{u}\|_{W^{1,\gamma}(\Omega)}, \|\mathbf{u}_h\|_{W^{1,\gamma}(\Omega)}, \|\mathbf{v}_h\|_{W^{1,\gamma}(\Omega)})$. The desired result (4.14a) follows from (4.15) with $s = r$, (2.23), (b), (d) and (3.6a) if we can show that

$$|\mathbf{u} - \pi_h\mathbf{u}|_{(r,\alpha)} \le C\left(\|\mathbf{u}\|_{C^{2,(2-\kappa)/\kappa}(\overline{\Omega})}, \|\mathbf{u}\|_{W^{3,1}(\Omega)}\right) h\ . \tag{4.16}$$

From (4.11), we have that

$$|D(\mathbf{u} - \pi_h\mathbf{u})(\mathbf{x})| \le Ch \sup_{y\in\Delta_\tau} |H(\mathbf{u})(\mathbf{y})| \quad \forall \mathbf{x} \in \overline{\tau}\ , \tag{4.17}$$

where $|H(\mathbf{u})|^2 = |H(u_1)|^2 + |H(u_2)|^2$ and $H(v)$ is the Hessian matrix of v. As $\mathbf{u} \in [C^{2,(2-\kappa)/\kappa}(\overline{\Omega})]^2$, it follows that

$$|D(\mathbf{u} - \pi_h\mathbf{u})(\mathbf{x})| \le Ch|H(\mathbf{u})(\mathbf{x})| + \hat{C}\left(\|\mathbf{u}\|_{C^{2,(2-\kappa)/\kappa}(\overline{\Omega})}\right) h^{2/\kappa} \quad \forall \mathbf{x} \in \overline{\tau}\ . \tag{4.18}$$

It is easy to check that the funtion $\psi(t) \equiv (a + t)^\sigma t^2$ with $a \ge 0$ and $\sigma \in (-1, 0]$ is increasing on $\mathbb{R}^+$ and that $\psi(|t_2 + t_2|) \le 2[\psi(|t_1|) + \psi(|t_2|)]$ for all $t_1, t_2 \in \mathbb{R}$. Thus we have from (3.1), (4.18) and the above that

$$\begin{aligned}
|\mathbf{u} - \pi_h\mathbf{u}|_{(r,\alpha)} \le & \int_\Omega \left(|D(\mathbf{u})| + |D(\mathbf{u} - \pi_h\mathbf{u})|\right)^{\alpha(r-2)} |D(\mathbf{u} - \pi_h\mathbf{u})|^2 \\
&\le Ch^2 \int_\Omega (|D(\mathbf{u})| + Ch|H(\mathbf{u})|)^{\alpha(r-2)} |H(\mathbf{u})|^2 \\
&\quad + \hat{C}h^{4/\kappa} \int_\Omega \left(|D(\mathbf{u})| + \hat{C}h^{2/\kappa}\right)^{\alpha(r-2)} \\
&\le C\left(\|\mathbf{u}\|_{C^{2,\alpha}(\overline{\Omega})}\right) h^2 \left[1 + \int_\Omega |D(\mathbf{u})|^{\alpha(r-2)} |H(\mathbf{u})|^2\right]\ . \qquad (4.19)
\end{aligned}$$

Next we note that for $\sigma \in (-1, 0]$

$$\int_\Omega |D(\mathbf{u})|^\sigma |H(\mathbf{u})|^2 \le C \sum_{i=1}^{3} \int_\Omega |w_i|^\sigma |\nabla w_i|^2\ , \tag{4.20}$$

where $w_1 \equiv \partial u_1/\partial x_1$, $w_2 \equiv \partial u_1/\partial x_2 + \partial u_2/\partial x_1$ and $w_3 \equiv \partial u_2/\partial x_2$. Furthermore, by Green's formula with $(n_1, n_2)^{\mathrm{T}}$ the outward normal of Γ, we have that

$$\begin{aligned}
(\sigma + 1) \int_\Omega |g|^\sigma |\partial g/\partial x_i|^2 = & \int_\Gamma \text{sign } (g)|g|^{\sigma+1} n_i \partial g/\partial x_i \\
& - \int_\Omega \text{sign } (g)|g|^{\sigma+1} (\partial^2 g/\partial x_i^2) \\
& \le C\|g\|^{\sigma+1}_{C(\overline{\Omega})} \|g\|_{W^{2,1}(\Omega)}\ , \quad i = 1 \text{ and } 2\ , \qquad (4.21)
\end{aligned}$$

where we have applied a trace inequality. Hence the desired result (4.16) follows from (4.19), (4.20) and (4.21).

The result (4.14b) follows from (4.14a), (4.15), (4.16) and applying the technique in the proof of Theorem 4.2(i) with $\xi = 2$ yielding $s = 2$. □

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